جامعة الأهرام الكندية
AHRAM CANADIAN UNIVERSITY

Faculty of computer science and information technology

Academic Year 2019/2020

Level:1
Course Name: Calculus 1
Course Code: MAT102

Sheet No.:2
Semester: Fall 2019
Instructor: Dr. RawhyIsmail

Problem 1: Sketch the graph of $y = e^x$. Clarify its domain and range. Find $lim_{x\to\infty} e^x$ and $lim_{x\to-\infty} e^x$. Repeat for $y = e^{-x}$, $y = 4^x$, and $y = (1/8)^x$.

Problem 2: Find the domain of the following functions:

(a) $\mathbf{y = e^{\frac{1}{x(x-1)}}}$

(b) $\mathbf{y = e^{2x^2-3x-1}}$

(c) $\mathbf{y = e^{\sqrt{9-x^2}}}$

Problem 3: Find the derivative of the following functions:

(a) $\mathbf{y = e^{2x^2-6\sqrt{x}+1}}$

(b) $\mathbf{y = e^{\frac{1-\sqrt{x}}{1+\sqrt{x}}}}$

(c) $\mathbf{y = 8^{1/x-6x+1}}$

(d) $\mathbf{y = \pi^x \quad x^\pi}$

(e) $\mathbf{y = 10^{\sqrt{x}-\frac{1}{x}} + (\sqrt{x} - \frac{1}{x})^{10}}$

(f) $\mathbf{y = (2x-3)^e \quad e^{2x-3}}$

Problem 4: Evaluate the following expressions:

(a) $\mathbf{\ln \sqrt[a]{e}}$ =............

(b) $\mathbf{\ln \frac{1}{e^{\sqrt{a}}}}$ =............

(c) $\mathbf{\ln(-e)}$ = ...

(d) $\mathbf{\ln(e\sqrt{e})}$ = ...

Problem 5: Solve the following equations

(a) $2e^{-x} - 3 = 10$

(b) $\sqrt{e^{1-3x}} = 2$

(c) $\frac{e^{2x}+11}{e^{2x}+1} = 4$

(d) $5 + 4\ln\sqrt{x} = 10$

(e) $\ln(3-4x) = 1$

(f) $\ln\left(\frac{1+x}{1-x}\right) = 4$

Problem6: Find the domain of the following functions:

(a) $y = \ln(4-3x)$

(b) $y = \ln(x^2+1)$

(c) $y = \ln\left(\frac{x}{x-1}\right)$

Problem 7: Find the drevative of the following functions:

(a) $y = \ln(1-10x^4)$

(b) $y = \sqrt{ln\left(5x - \sqrt{x} + \frac{1}{x}\right)}$

*(c) $y = e^{\ln(10x)}$

(d) $y = 4^{\ln(5x-1)}$

(e) $y = \sqrt[4]{\frac{\left(2\sqrt{x} - \frac{4}{x}+1\right)}{e^{-x^2}\left(\frac{10}{x^2}-6\sqrt{x}+1\right)}}$